

THE 1ST COASTAL OCEAN ENVIRONMENT SCHOOL IN WESTERN INDIAN OCEAN (COES-WIO)



MOMBASA, KENYA

21ST - 27TH SEPTEMBER 2025



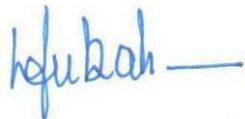
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FOREWORD

Technical University of Mombasa (TUM) in Kenya hosted the inaugural Coastal Ocean Environment School in the Western Indian Ocean (COES-WIO) 2025, a pioneering initiative that brought together science, training, and regional collaboration in support of sustainable ocean science and governance. The WIO region comprising ten states is rich in biodiversity and marine resources supporting millions of people directly and indirectly. Coastal communities are increasingly vulnerable to the impacts of climate change, declining fisheries resources, marine pollution, and ecosystem degradation. At the same time, the region holds immense potential within the blue economy, provided that investments in knowledge, capacity, and innovation are strengthened.

COES-WIO represents a strategic response to this need. By bringing together participants from across the WIO region and beyond, the training created a platform for hands-on learning, interdisciplinary engagement, and knowledge exchange grounded in real-world coastal environments. This training aligned with TUM's growing role in education, research, innovation, and global ocean science engagement. This places TUM and Kenya at the centre particularly Kenya hosting the UN Environment Program (UNEP) and the Nairobi Convention aimed at protecting, managing, and developing the marine and coastal environment of the WIO region. We remain committed to supporting and scaling up initiatives such as the COES-WIO that contribute to building a skilled, informed, and collaborative ocean science community in the WIO region and Africa at large.



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VICE CHANCELLOR

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ACKNOWLEDGEMENT

The successful role out of COES-WIO 2025 was made possible through the strong collaboration between academic institutions, research organizations, regional and global partners. We acknowledge the financial support from the US National Science Foundation (Grant No. OCE-2319142) through the University of Michigan (UoM) to Technical University of Mombasa (TUM) which seamlessly facilitated the implementation of the training. We also thank the Ocean Decade Capacity Development Facility, and the Ocean Teacher Global Academy (OTGA) who supported some participants in the training through travel facilitation. We extend our sincere appreciation to our partner institutions who played a big role in the planning and execution of the training. These are: Kenya Marine and Fisheries Research Institute (KMFRI), Pwani University (PU), Western Indian Ocean Marine Science Association (WIOMSA), and the Early Career Ocean Professionals (ECOP) Programme. We also recognize the invaluable contributions of both local and international instructors, facilitators, technical teams, and the local organizing committee. Finally, we thank all participants for their enthusiasm, engagement, and commitment to advancing ocean science in the WIO region.

EXECUTIVE SUMMARY

The Coastal Ocean Environment School in Western Indian Ocean (COES-WIO) is a summer school initiated by the University of Michigan, USA. For several years before COES-WIO, the University of Michigan has been conducting similar summer schools in West Africa (Ghana and Nigeria) and this initiative has been embraced in the WIO region. The 1st COES-WIO 2025 school was co-hosted in Kenya, by Technical University of Mombasa (TUM), Pwani University and the Kenya Marine and Fisheries Research Institute (KMFRI). The successful implementation of this school was as a result of a dedicated planning process which started way back in the year 2023 led by Prof. Brian Arbic of the University of Michigan, in consultation with local partners at TUM, Pwani University and KMFRI. Initially meant to be a Kenya school targeting only Kenyan participants, this initiative attracted other partners including the Western Indian Ocean Marine Science Association (WIOMSA), IOC UNESCO and Early Career Ocean Professionals (ECOP). As a result of this involvement with partners in the WIO region, COES-WIO has become a regional school drawing participants and resource persons across the WIO region and beyond with the following objectives:

- To enhance regional capacity in ocean science, especially among early-career professionals;
- To provide hands-on, practical training in key thematic areas of coastal and marine environmental research;
- To foster interdisciplinary collaboration and networking among ocean scientists, researchers, and policymakers across Africa;
- To promote the exchange of knowledge and best practices in sustainable marine resource management,
- To attract professionals from across the globe to work in partnership with the rich network in the WIO region, and
- To strengthen partnerships among academic institutions and research organizations within the WIO region.

The 1st COES-WIO 2025 summer school had 5 hands-on project training tracks: Small-scale Fisheries, Mariculture and Post-harvest Management; Chemical and Physical Oceanography; Climate Change and Physical Processes; Marine pollution and Plastics, and Ecological Survey Techniques and Data Management. These project tracks were designed and facilitated by both regional and international subject matter experts who were carefully identified during the planning phase. The project training tracks were selected to build the needed capacity in the WIO region amongst early career professionals, post-graduate and undergraduate participants. These training tracks in the region are supposed to encourage participants to develop research interest in the related selected disciplines. Training sessions were conducted through interactive classroom lectures, group exercises, field excursions and laboratory practical exercises, and oral presentations of findings. Selected field sites in the north and south coast of Kenya offered exciting field work experiences to all the participants

and instructors. The 1st COES-WIO summer school was also officially endorsed as a pre-WIOMSA Scientific Symposium event. Many participants attended the 13th WIOMSA Scientific Symposium which was also held in Mombasa the following week after the summer school.

The week-long summer school trained a total of 75 participants drawn across the WIO region and beyond with a record of 17 countries represented. Three major social events which will remain memorable to both participants and facilitators were the Welcoming Dinner that was officiated by Prof. Laila Abubakar (TUM Vice Chancellor), the Opening and Closing Ceremonies with very exciting speeches from the organizing institutions' heads and directors. These memorable social events included the irresistible Mijikenda traditional dance organized by members of TUM staff. The 1st COES-WIO summer school was a resounding success, which met its overarching objectives. The program created a vibrant platform for learning, exchange, and professional development among the WIO region early career ocean scientists, postgraduate and undergraduate students. Plans are now underway for the 2nd COES-WIO 2026 summer school to be held in Tanzania.

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LIST OF ABBREVIATIONS AND ACRONYMS

CBO	- Community-Based Organization
CNRS	- Centre National de la Recherche Scientifique
COES-WIO	- Coastal Ocean Environment School in the Western Indian Ocean
COESSING	- Coastal Ocean Environment Summer School in Nigeria and Ghana
ECOP	- Early Career Ocean Professionals
GPS	- Global Positioning System
IBA	- Important Bird Area
IMTA	- Integrated Multi-Trophic Aquaculture
JAMSTEC	- Japan Agency for Marine-Earth Science and Technology
KMFRI	- Kenya Marine and Fisheries Research Institute
MCCCCG	- Mida Creek Community Conservation Group
pH	- Potential Hydrogen
PU	- Pwani University
SSFs	- Small-Scale Fisheries
TSS	- Total Suspended Solids
TUM	- Technical University of Mombasa
UDOM	- University of Dodoma
UM	- University of Michigan
WIOMSA	- Western Indian Ocean Marine Science Association
WIO	- Western Indian Ocean
WIO-ECSN	- Western Indian Ocean Early Career Scientists Network

CHAPTER ONE

1.0 INTRODUCTION

1.1 Background of COES-WIO

The inaugural Coastal Ocean Environment School in the Western Indian Ocean (COES-WIO) was successfully held at the Technical University of Mombasa (TUM), Kenya, from 21st to 27th September 2025. This first-of-its-kind regional training brought together over 100 participants from 18 African countries of the Western Indian Ocean (WIO) region. This training marked a significant milestone in strengthening regional collaboration, knowledge exchange, and technical capacity in ocean science and coastal environment management across the WION region. The COES-WIO 2025 training was implemented through a strategic partnership between the Technical University of Mombasa (TUM) and the University of Michigan (UoM), USA, following the successful award of a joint grant.

This collaboration combined local expertise and regional knowledge with international scientific experience, ensuring both contextual relevance and technical depth. The program was further strengthened through the support of key regional and international institutions, including the Kenya Marine and Fisheries Research Institute (KMFRI), Pwani University, the Early Career Ocean Professionals (ECOP) network, and the Western Indian Ocean Marine Science Association (WIOMSA). These partnerships played a critical role in facilitating training delivery, access to field sites, technical mentorship, and broader regional engagement. The call for applications was open for a period of 15 days, from 1st to 15th July 2025. While the program initially targeted 75 participants from the WIO region, it received an overwhelming 812 applications from over 45 countries globally. This strong response highlights the significant unmet demand for high-quality, practical training in ocean science, particularly within the WIO region where access to such opportunities remains limited. The applicant pool reflected a diverse range of academic backgrounds and professional interests, with 39% female and 61% male applicants. The majority held Bachelor's degrees, with strong representation at Master's and PhD levels, while less than 5% were undergraduates.

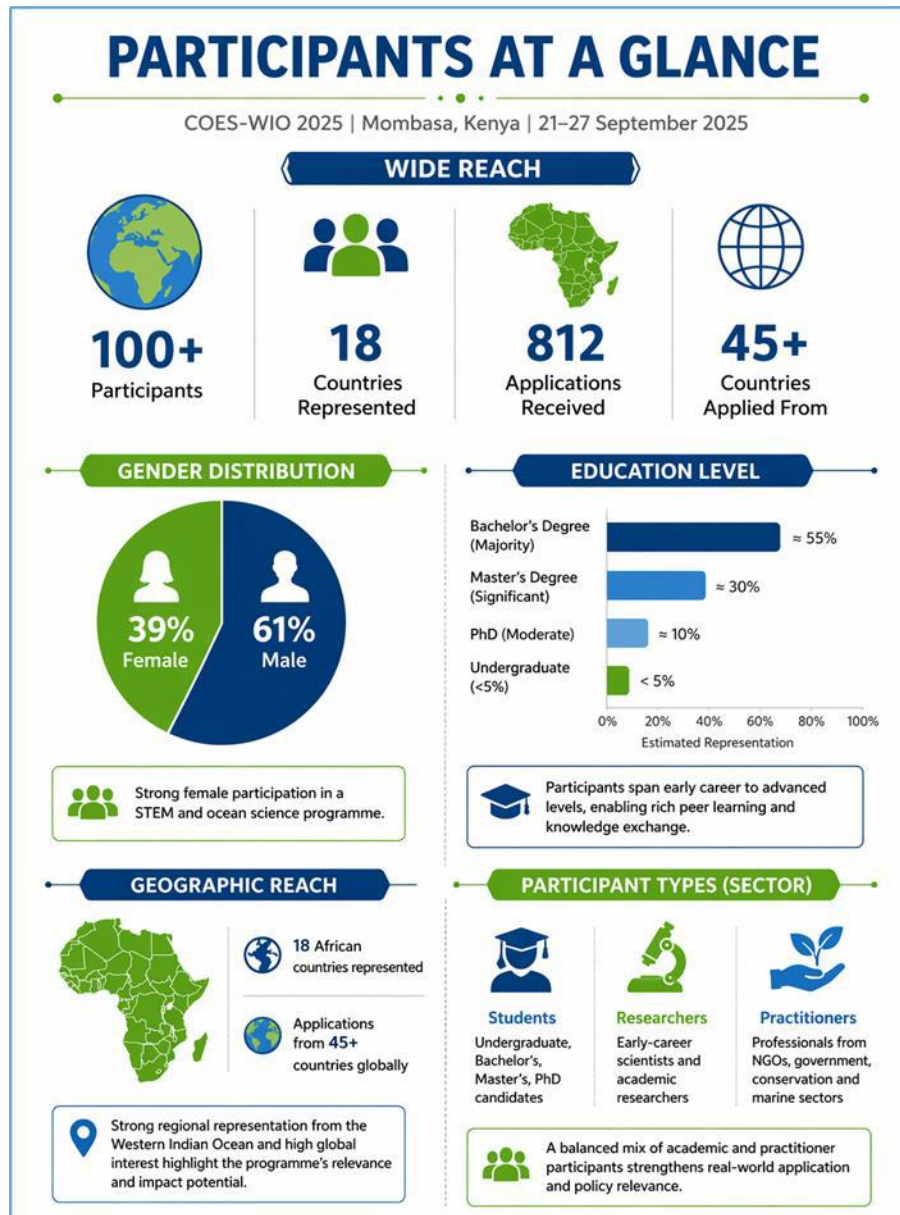


Figure 1. Details of COES-WIO 2025 participants demography

The participant selection process was highly competitive and carefully structured to ensure both academic excellence and regional representation. Applications were grouped according to the five thematic project tracks, based on each applicant's area of interest. Instructors for each track conducted detailed evaluations, selecting 15 participants per track based on quality, relevance, and potential impact. An additional five candidates per track were placed on a waiting list. The final cohort was intentionally balanced to reflect diversity in gender, academic level, and geographic origin across the WIO region, creating an inclusive and interdisciplinary learning environment that fostered peer learning, collaboration, and cross-country exchange of knowledge and experiences.

Overall, the successful delivery of the inaugural COES-WIO demonstrates both the feasibility and the strong demand for regionally anchored, high-impact training programs in ocean science within the WIO. By building on a proven continental model and adapting it to the WIO context, COES-WIO has laid a strong foundation for sustained capacity development, strengthened regional collaboration, and the emergence of a new generation of ocean science professionals equipped to support sustainable management of marine and coastal resources in the region. Figure 2 below shows the global distribution of the applicants.

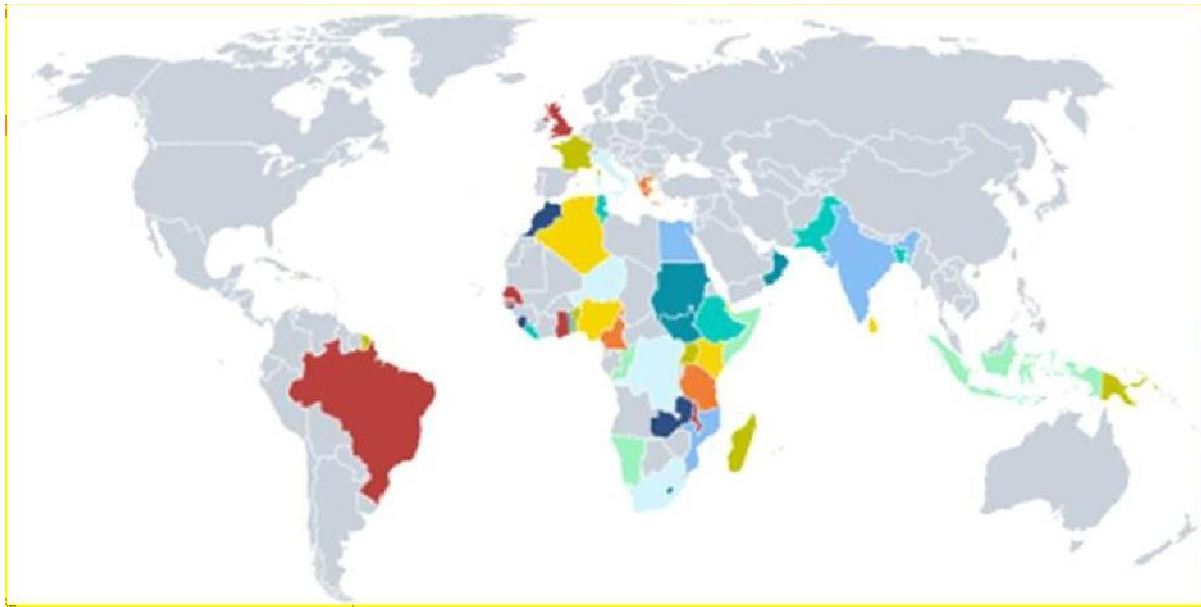


Figure 2. Global map showing the distribution of applicants

1.2 Western Indian Ocean Context

The Western Indian Ocean (WIO) region comprises a diverse group of coastal and island nations including Kenya, Tanzania, Mozambique, Somalia, South Africa (eastern coast), Madagascar, Comoros, Seychelles, Mauritius and Reunion. Collectively, these countries are connected by shared marine ecosystems, socio-economic dependencies, and transboundary ocean processes.

The region is characterized by an extensive coastline estimated at over 20,000 km, supporting a rapidly growing coastal population exceeding 60 million people (UNEP-Nairobi Convention and WIOMSA, 2015). These communities rely heavily on coastal and marine resources for their livelihoods, food security, and economic development. Coastal and ocean-based sectors including fisheries, aquaculture, tourism, shipping and emerging blue economy industries play a central role in national and regional economies.

Ecologically, the WIO is one of the most biodiverse marine regions globally. It hosts critical ecosystems such as coral reefs, mangrove forests, seagrass beds, estuaries, and

coastal wetlands. These ecosystems provide essential services, including shoreline protection, carbon storage, habitat provision, and support for fisheries. The region is also recognized for its high levels of endemism and its importance as a habitat for migratory species, including fish, marine mammals, and seabirds (UNEP-Nairobi Convention and WIOMSA, 2015). However, the sustainability of these ecosystems and the livelihoods they support is increasingly challenged by pressures such as overexploitation of marine resources, marine pollution, and rapid coastal development. These challenges highlight the urgent need for strengthened scientific capacity, improved monitoring systems, and enhanced regional collaboration to support sustainable ocean management.

Despite the strategic importance of the WIO region, access to advanced training, research infrastructure, and interdisciplinary learning opportunities remains limited. This capacity gap constrains the ability of researchers, practitioners, and institutions to generate and apply scientific knowledge effectively. In this context, initiatives such as COES-WIO are critical in building the technical expertise and collaborative networks needed to support sustainable management and long-term development of the region's marine and coastal resources.

1.3 History and Development of COES-WIO

The establishment of COES-WIO builds on a well-established and successful model of ocean science training in Africa. The initiative was led by Brian Arbic, with strong institutional and regional support from Cosmas Munga. Prof. Arbic is the founder of the Coastal Ocean Environment Summer School (COESSING), a long-standing training program that has been implemented for over a decade across West Africa, particularly in Nigeria and Ghana.

COESSING has played a critical role in strengthening ocean science capacity in West Africa by providing hands-on, interdisciplinary training to early-career scientists and practitioners. The success of this model demonstrated the value of regionally anchored training programs that combine theory with practical application. COES-WIO represents a strategic expansion of this proven model to the WIO region. The program has been adapted to reflect the unique environmental conditions, institutional landscape, and development priorities of the WIO, while maintaining its core emphasis on experiential learning, interdisciplinary engagement, and regional collaboration. As such, COES-WIO provides a platform for building a new generation of ocean science professionals equipped to address coastal and marine challenges within the region.

1.4 Objectives

The main objective of COES-WIO 2025 was to build technical and analytical capacity in ocean and coastal sciences through an integrated and experiential learning approach. The specific objectives were to:

- i. To enhance regional capacity in ocean science, especially among early-career professionals;

- ii. To provide hands-on, practical training in key thematic areas of coastal and marine environmental research;
- iii. To foster interdisciplinary collaboration and networking among ocean scientists, researchers and policymakers across Africa;
- iv. To promote the exchange of knowledge and best practices in sustainable marine resource management;
- v. To attract professionals from across the globe to work in partnership with the rich network in the WIO region, and
- vi. To strengthen partnerships among academic institutions and research organizations within the WIO region

1.4 Rationale of COES-WIO

Despite growing global attention to ocean science, the WIO region continues to face a critical gap in technical capacity. Limited access to advanced training, research infrastructure, and interdisciplinary learning opportunities constrains the ability of researchers, practitioners, and institutions to effectively address complex coastal and marine challenges.

COES-WIO was established to address this gap by providing a practical, interdisciplinary platform for hands-on training, knowledge exchange, and regional collaboration. By equipping participants with relevant technical skills and fostering cross-sector engagement, the program strengthens the foundation for locally driven, science-based solutions. COES-WIO aligns with global and regional priorities, including the UN Ocean Decade and supports sustainable ocean management and blue economy development in the WIO region.

CHAPTER TWO

2.0 METHODOLOGY AND PROGRAMME DESIGN

The COES-WIO 2025 program was designed as an integrated, interdisciplinary training platform aimed at delivering both theoretical knowledge and practical skills in coastal and marine sciences. The methodology combined a rigorous participant selection process, a structured multi-component training approach, and the use of diverse coastal field sites to ensure contextualized, hands-on learning. The overall design emphasized experiential learning, regional relevance, and the application of scientific knowledge to real-world coastal and marine challenges in the WIO region.

2.1 Participant Selection

The call for applications was open from 1st to 15th of July 2025 and attracted an overwhelming 812 applications from over 45 countries worldwide. This strong response reflected both the high demand for capacity-building opportunities in ocean science and the growing recognition of the importance of interdisciplinary training in addressing coastal and marine challenges.

The selection process was highly competitive and structured to ensure transparency, quality, and diversity. Applicants were required to indicate their preferred thematic track and demonstrate their academic background, motivation, and relevance of their work or interests to the program. Applications were then grouped according to the five thematic project tracks and reviewed by the respective track instructors.

Selection criteria included:

- Academic qualifications and disciplinary relevance;
- Demonstrated interest or experience in ocean and coastal sciences;
- Motivation and potential for impact in the applicant's home country or institution, and
- Alignment with the selected thematic track.

In addition to academic merit, deliberate efforts were made to ensure:

- **Gender balance**, promoting participation of women in ocean science
- **Regional representation**, with a focus on inclusivity across the WIO region
- **Diversity in academic levels**, including Bachelor's, Master's, and PhD participants

Each thematic track selected 15 participants, with an additional 5 participants placed on a waiting list to accommodate possible withdrawals. This structured approach ensured manageable group sizes for effective learning while maintaining high-quality engagement between participants and instructors.

2.2 Programme Structure

The COES-WIO training programme was designed as a comprehensive and integrated learning experience, combining four key components that reinforced each other to enhance participant understanding and skill development.

2.2.1 Classroom-Based Theoretical Sessions

Classroom sessions provided foundational knowledge in the following key thematic areas:

- i. Climate change and shoreline change processes
- ii. Ecological survey techniques and data management
- iii. Marine plastics and pollution
- iv. Physical and chemical oceanography
- v. Small-scale fisheries and aquaculture

These sessions introduced core concepts, theoretical frameworks, and analytical approaches that guided subsequent practical activities.

2.2.2 Field-Based Experiential Learning

Fieldwork formed a central component of the program, allowing participants to directly engage with real-world coastal environments. Participants conducted *in-situ* observations, collected environmental data, and interacted with the local communities and practitioners. This approach enabled participants to contextualize theoretical knowledge and understand the complexity of coastal and marine systems in practice.

2.2.3 Laboratory Analysis

Laboratory sessions provided hands-on training in analytical techniques used in ocean science. Participants processed samples collected during fieldwork, including fish post-harvest techniques and value addition, water quality parameters and microplastics, using methods such as filtration, microscopy, and spectrophotometry. These activities strengthened participants' technical competencies and reinforced the link between field data and scientific analysis.

2.2.4 Data Analysis and Modelling

Participants were introduced to data processing, visualization, and interpretation using appropriate tools and software. This included basic statistical analysis, interpretation of environmental datasets, and exposure to modelling approaches. The integration of data analysis enhanced participants' ability to translate observations into meaningful scientific insights.

The combination of these four components ensured a holistic learning experience that moved from theory to practice and from data collection to interpretation.

2.3 Study Sites

Field activities were conducted across a range of strategically selected coastal sites along the Kenyan coastline, each representing distinct ecological characteristics and varying levels of human influence. These sites were chosen to expose participants to the diversity of coastal systems within the WIO region and to illustrate the interactions between natural processes and human activities.

2.3.1 Gazi Bay

Gazi Bay in south coast Kenya is a well-studied estuarine system characterized by mangrove forests, seagrass beds, and adjacent coral reef ecosystems. The site provided an ideal setting for studying biophysical gradients, water quality dynamics, and ecosystem interactions. It also enabled participants to observe the role of coastal ecosystems in supporting biodiversity and local livelihoods.

2.3.2 Kilifi Creek

Kilifi Creek in north coast Kenya, particularly at Kibokoni area represents a mangrove-dominated coastal system where community-based aquaculture and fisheries activities are practiced. The site offered insights into integrated aquaculture systems and the relationship between environmental conditions and livelihood strategies.

2.3.3 Mida Creek

Mida Creek in north coast Kenya, particularly at Dabaso area is an ecologically significant area known for its mangrove ecosystems, biodiversity, and community-based conservation initiatives. It provided a platform for understanding ecosystem services, eco-tourism, and sustainable resource management practices.

2.3.4 Diani Beach

Diani Beach in south coast Kenya is a major tourist destination with relatively well-managed coastal environments. It served as a comparative site for studying marine pollution, particularly microplastics, in areas with controlled human activity.

2.3.5 Pirates Beach

Pirates Beach in Mombasa, is a peri-urban coastal site characterized by high human activity and visible signs of shoreline change. It provided an opportunity to examine coastal dynamics, erosion processes, and the impacts of human infrastructure on shoreline stability.

2.3.6 Rationale for Site Selection

The selection of these sites was guided by the need to:

- Represent a range of coastal ecosystems (mangroves, seagrass, coral reefs, sandy beaches);

- Capture varying degrees of anthropogenic influence;
- Provide opportunities for interdisciplinary learning across thematic tracks, and
- Enable direct engagement with local communities and real-world challenges

By working across these diverse environments, participants gained a comprehensive understanding of coastal systems and the factors influencing their sustainability.

CHAPTER THREE

3.0 PROGRAM DELIVERY

3.1 Opening and Closing Ceremonies

The COES-WIO program commenced and concluded with formal opening and closing ceremonies, which played an important role in setting the tone for the training and celebrating its outcomes. These events brought together institutional leaders, partners, instructors, and participants, reinforcing the collaborative nature of the program (Figure 3).



Figure 3. Speeches from heads of partner institutions, top-left: Prof. Laila Abubakar (TUM); top-right: Prof. James Kahindi (PU); bottom-right: Prof. Brian Arbic (UoM), and bottom-right: Dr. Jacob Ochiewo (KMFRI)

The opening ceremony was officiated by Prof. Laila Abubakar, the Vice Chancellor of the host institution, Technical University of Mombasa (TUM), who welcomed participants and emphasized the importance of strengthening ocean science capacity in the region. Remarks from organizing partners and institutional representatives highlighted the

significance of regional collaboration and the role of training in advancing ocean research and sustainable management.

Social engagement was integrated into the programme through a welcoming dinner and a farewell dinner, both held at the TUM Hotel and Hospitality Centre. These events provided opportunities for networking, informal interaction, and cultural exchange among participants and facilitators (Figure 4).



Figure 4. Opening ceremony of the COES WIO 2025 at Technical University of Mombasa

A highlight of the social program was a traditional coastal Mijikenda dance performance organized by TUM staff, which showcased local culture and enriched the overall participant experience. The closing ceremony provided an opportunity to reflect on the week's activities, acknowledge achievements, and reinforce connections established during the program.

3.2 Training Tracks

3.2.1 Small-scale Fisheries, Mariculture and Post-Harvest Management

The course work was conducted at Technical University of Mombasa, and the field work was conducted at Kibokoni and Dabaso at Mida Creek in north coast Kenya (Figure 5). This training track: small-scale fisheries, mariculture and post-harvest management was facilitated by the following experts:

- Prof. Cosmas Munga - Technical University of Mombasa, Kenya
- Dr. Anthony Nzioka - Kenya Marine and Fisheries Research Institute
- Dr. David Mirera - Kenya Marine and Fisheries Research Institute
- Dr. Joseph Tunje - Pwani University, Kenya
- Dr. Peter Odote - Kenya Marine and Fisheries Research Institute
- Mr. Frank Mirobo - University of Dodoma, Tanzania



Figure 5. Experts of the training track and a field session at Kibokoni area, north coast Kenya

This training track had 15 students who had interest in studying fisheries, mariculture and postharvest management. During fieldwork, participants visited Kibokoni community mariculture project situated along Kilifi Creek. The Kibokoni community is undertaking an Integrated Multitrophic Aquaculture (IMTA) project where different aquatic marine organisms at different trophic levels are farmed in one system. Apart from learning the general benefits of aquaculture, participants also learnt the economic and ecological benefits of such a project, and participated in a practical session of fish harvesting, and fish data recording (weight and length).

The Crab Shark Project in Dabaso, Mida Creek was the second site to be visited. This site falls within Mida Creek where a community-based organization (CBO), Mida Creek Conservation Group (MCCG) is carrying out mariculture, crab eco-tourism, and mangrove restoration activities. In mariculture, the group is undertaking crab fattening using cages. At Dabaso, a member of the CBO gave a presentation on crab farming, and participants had the chance of asking questions. The students also participated in crab harvesting, and took the necessary measurements, that is crab weight, length, etc.

Participants were introduced to the importance of small-scale fisheries (SSFs) in supporting livelihoods, income, and food security in the WIO region. The training provided an integrated overview of SSFs and aquaculture, focusing on production, sustainability, and post-harvest management in the Kenyan context and the WIO at large (Figure 6).



Figure 6. Participants in the small-scale fisheries, mariculture and post-harvest management training in a practical session at Kibokoni, north coast Kenya

Under small-scale fisheries, the learners were taken through fisheries potential in Kenya, current status, challenges and threats as well as existing management strategies. In groups, the learners also described and grouped various fish species by habitat types and trophic level, using the FishBase. Under mariculture, the students were taken through global and regional mariculture production statistics, mariculture classification, culture systems and species and seaweed farming systems. Participants were also taken through hands-on practical sessions on finfish culture techniques, crustacean culture techniques, live feed production and seaweed cultivation. Under post-harvest management, the participants were taken through post-harvest intervention, status of fish landing sites and value chain improvement, using the Kenyan coast as a case study (Figure 7).



Figure 7. Participants in a practical session on fish value addition at KMFRI lab, Mombasa

3.2.1.1 Description of Training Sites

Participants in the training track of small-scale fisheries, mariculture and postharvest management visited Kibokoni and Dabaso community aquaculture projects for their field work and practical sessions. Below is a brief description of these study sites.

Kibokoni is a small rural village along Kilifi Creek, north coast Kenya. Geographically, the village is located at latitude 3° 36' 7''S and longitude 39° 48' 8'' E, and at an altitude of 20 m above the sea level. It is about 6.5 km west of Kilifi Town (Figure 8).

Biophysically, Kibokoni area is generally warm throughout the year with temperatures ranging between 21°C during the coldest month (June and July) and 32°C during the hottest months (January and February), with annual rainfall amounts up to 1300 mm which is bimodal in pattern – the long rains come in May while the short rains are experienced in the months of October.



Figure 8. Mangroves at Kibokoni along Kilifi Creek in north coast Kenya

Kibokoni village, being adjacent to Kilifi Creek, is endowed with mangrove forest. The creek is home to six mangrove species: *Avicennia marina*, *Ceriops tagal*, *Sonneratia alba*, *Rhizophora mucronata*, *Lumnitzera racemosa* and *Bruguiera gymnorhiza*. The most dominant mangrove species in the area is *Avicenna marina* with a percentage stand of 25.6%, while the less dominant species of *Lumnitzera racemosa* and *Heritiera littoralis* has a stand of 0.10% (Said, 2016).

Kibokoni village had 473 households with a population of 3,336 people, out of which 30.5% were males, 43.0% were females, and 26.6% were children below 5 years of age. Farming and fishing are the main livelihood activities. Due to decline in fish catches, and through the support of government and non-governmental agencies, the community is practicing aquaculture. During the field visit Kibokoni had an integrated multi-trophic aquaculture (IMTA) project where different aquatic species (prawns, crab, milkfish and tilapia) were being cultured together in ponds. Here, participants interacted with the locals and learnt about some aspects of aquaculture (Figure 9).



Figure 9. Participants during a practical session of fish harvesting at Kibokoni, north coast Kenya

Dabaso village is found along Mida Creek and at an altitude of 3.35°S and longitude 40.02° East, about 4.5 km west of Watamu Town. The village, where Crab Shark Project is domiciled, is found on the eastern side of Mida Creek, with a low-lying and gentle undulating topography, and elevation which is generally close to the sea level. The area has a tropical climate with mean temperatures ranging between 24 -32°C, and receives rainfall amounts ranging between 1000-12000 mm per annum, the short rains come in March while the long rains are experienced in the month of October.

Mangrove forests dominate the intertidal zone, with the major species being *Rhizophora mucronate*, *Avicennia marina*, *Ceriops tagal*, and *Sonneratia alba*. The Mida Creek is internationally known for migratory birds, and has the status of a globally Important Birds Area (IBA). The area is environmentally sensitive and an important fishery, bio-diversity and blue carbon hotspot (Figure 10).

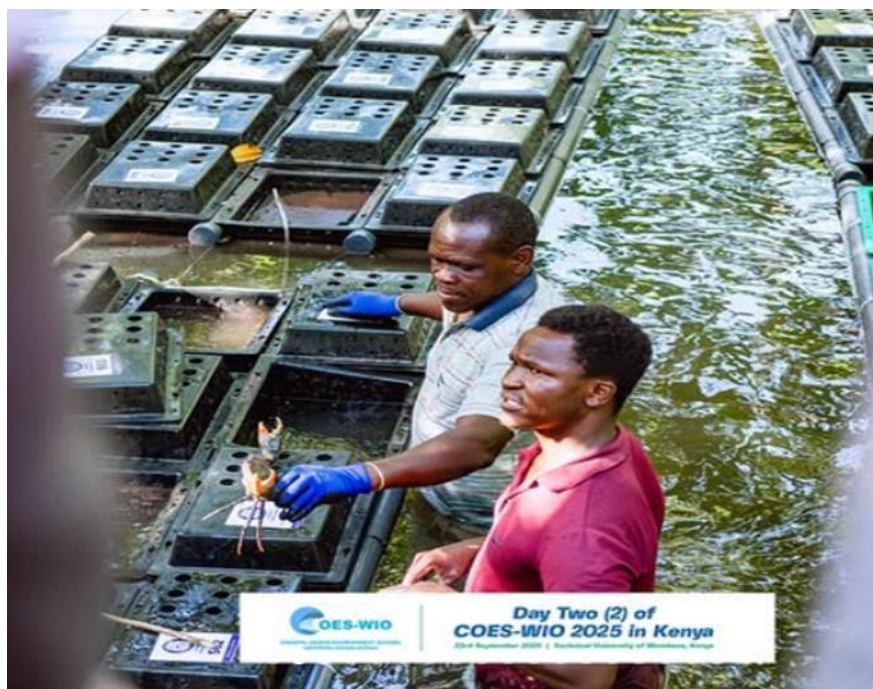


Figure 10. Crab Shark Project at Watamu for crab fattening

Tourism and fishing are the major livelihood activities in Dabaso village and Watamu in general. Part of the Mida Creek falls within the Watamu National Marine Park and Reserve. The Mida Creek Community Conservation Group (MCCCCG), is engaged in both environmental conservation and eco-tourism activities. The group runs the mangrove board walk and the floating crab shark restaurant, doubling up with aquaculture project where they do crab fattening.

As part of the Small-Scale Fisheries, Mariculture and Post-Harvest Management training track, participants visited Kibuyuni village in south coast Kenya to gain practical exposure to seaweed farming. The visit provided insights into cultivation practices, value chains, and the role of seaweed as an emerging livelihood activity in coastal communities. Seaweed farming was introduced as a form of mariculture involving macroalgae, classified into green, brown, red, and blue-green groups based on their pigments and structure. The training focused on two commonly farmed species: *Kappaphycus alvarezii*

(*cottonii*), which is more delicate, and *Eucheuma denticulatum* (*spinosum*), which is more resilient and adaptable to environmental conditions (Figure 11).



Figure 11. Seaweed harvesting at Kibuyuni, south coast Kenya

Participants were guided through key considerations for site selection, emphasizing that suitable sites should have sandy substrates, minimal freshwater intrusion, and moderate water movement. Areas with strong currents or excessive turbidity were identified as unsuitable, while shallow waters accessible during low tide were preferred to facilitate farming activities. Proper alignment of seaweed lines parallel to the shoreline was also highlighted as important for reducing resistance to tidal forces.

The field visit demonstrated the shallow water off-bottom cultivation method, which is widely practiced in the region. This method involves the use of ropes, wooden pegs, and tie materials to suspend seaweed seedlings in shallow coastal waters. Participants observed planting techniques where seed material, typically prepared in 100 g bundles, is tied at regular intervals along ropes stretched between pegs. Farming activities are conducted during low tide periods, allowing farmers to access and maintain the farms.

Harvesting practices were also demonstrated, with participants learning that seaweed typically matures within approximately six weeks (around 45 days). Early harvesting results in lower carrageenan content, affecting product quality. Harvesting involves manual removal of seaweed from the ropes, followed by washing to remove sand, debris, and epiphytes. Post-harvest processes includes drying, which is a critical step in maintaining product quality. Seaweed is dried on raised racks or designated platforms for 3–5 days under favourable weather conditions, preferably

under shade away from direct sunlight to preserve bioactive compounds. During rainy periods, drying may take up to seven days, and exposure to rain can negatively affect quality by causing bleaching or spoilage (Figure 12).



Figure 12. Seaweed drying at Kibuyuni, south coast Kenya

Participants were also introduced to basic processing and value addition processes, where dried raw seaweed is converted into flakes or powder using mechanical methods such as cutting, grinding, and milling. These processed products are used in a wide range of industries. Seaweed-derived hydrocolloids including agar, alginate, and carrageenan, are widely utilized in food processing, pharmaceuticals, and cosmetics. Additionally, seaweed is used directly in food products, personal care items such as toothpaste, soaps and lotions, and as a raw material for industrial applications (Figure 13). Overall, the field visit provided participants with a comprehensive understanding of seaweed farming as a sustainable livelihood option, highlighting its economic potential, low environmental impact, and relevance to coastal community development in south coast Kenya and the entire WIO region.



Figure 13. Seaweed products at Kibuyuni, south coast Kenya

3.2.2 Physical and Chemical Oceanography

This training track involved field observations of water stratification and mixing influencing habitat health. It also involved observation of winds, tides, river discharge, and waves that drive motion along coasts shaping up beaches, estuaries, and deltas.



Figure 14. Study sites at Gazi Bay, south coast Kenya



Figure 15. Participants of the Physical and Chemical Oceanography training track

The Physical and Chemical Oceanography training track was facilitated by the following experts:

- | | |
|----------------------------------|--|
| • Prof. Brian Arbic | - University of Michigan, USA |
| • Dr. Mariam Swaleh | - Technical University of Mombasa, Kenya |
| • Dr. Luis Fernando Pareja-Roman | - Rutgers University, USA |
| • Dr. Tracy Schafer | - University of Florida, USA |
| • Dr. Peter Knoop | - University of Michigan, USA |
| • Prof. Chenzhu Xu | - Oregon State University, USA |
| • Audrey Delpech | - French National Centre for Scientific Research, France |

3.2.2.1 Description of the Study Location

Fieldwork for the Chemical Oceanography training participants was conducted in Gazi Bay in south coast Kenya. Gazi Bay is a coastal estuarine system influenced by riverine inputs, tidal processes, and adjacent coastal ecosystems including mangroves, seagrass, and coral reefs (Figure 16).



Figure 18. Laboratory analysis of water samples

Students also conducted numerical modelling practical exercises to simulate salt intrusion and exchange flows (Figure 19). Satellite-derived sea surface temperature and salinity datasets were analysed to assess spatial and temporal variability (Figure 20). Engagement with local communities provided insights into human influences on estuarine systems.



Figure 19. Practical session modelling in computer laboratory at TUM

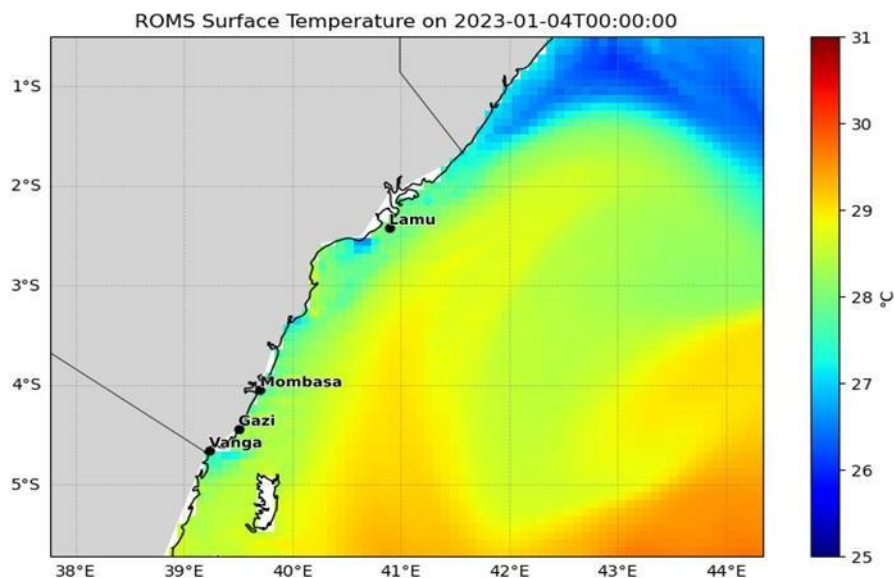


Figure 20. Results of sea surface temperature observation

Participants in this training track gained hands-on skills in field sampling, laboratory analysis, and numerical modelling. Findings showed strong variability in salinity, temperature, pH, and TSS driven by tidal cycles, freshwater inflow, and human activities (Figure 21). Increased turbidity and salt intrusion were found to influence mangrove, seagrass, and coral reef ecosystems.

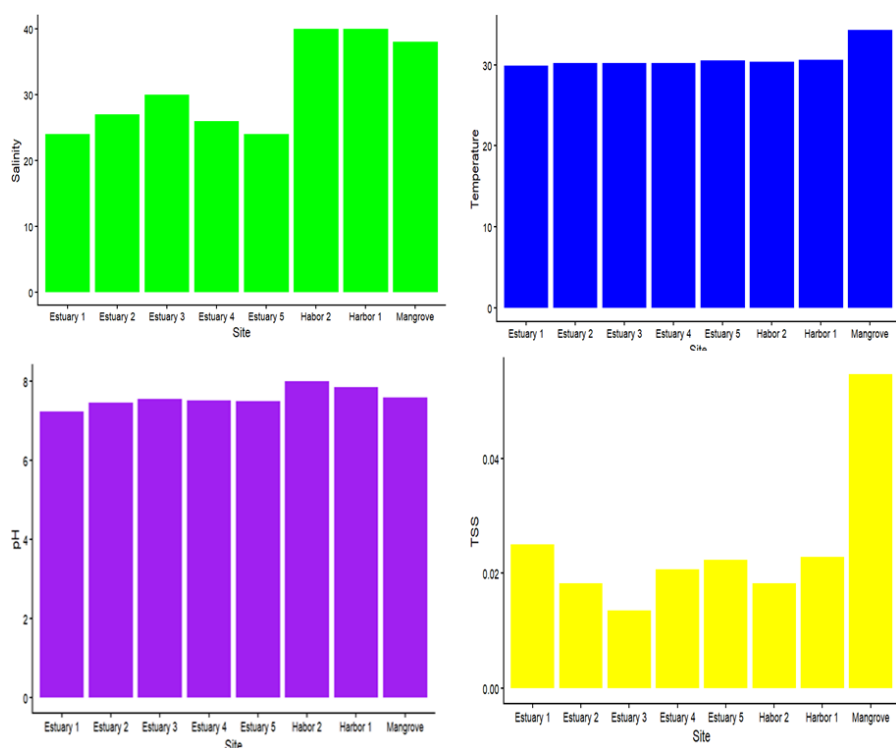


Figure 21. Results of physico-chemical parameters, Gazi Bay south coast Kenya

From this training track, the following were identified as the research implications:

- Monitoring these parameters is vital for sustainable coastal resource management.
- Results highlight the link between climate change, human activities, and ecosystem health in estuarine environments.
- Findings can guide policy makers, conservation managers, and local communities in mitigating impacts.
- Improve sampling frequency and spatial coverage.
- Integrate socio-economic data with physical and chemical oceanographic findings.
- Strengthen collaboration between research institutions and local stakeholders for long-term monitoring.

The Chemical Oceanography training track demonstrated how physical and chemical processes interact to shape estuarine environments. The integration of field data, laboratory measurements, and modelling enhanced understanding of coastal variability and highlighted the importance of sustained monitoring for effective coastal management.

3.2.3 Climate and Physical Processes

Participants in this training track participated in a blended programme combining classroom instruction, field observation, and data analysis. Field activities focused on systematic assessment of shoreline conditions, identification of erosion indicators, and evaluation of coastal infrastructure vulnerability. This training track was facilitated by the following experts:

- Prof. Ali Maarifa Mwakumanya- Pwani University
- Dr. Amon Kimeli - Kenya Marine and Fisheries Research Institute
- Mr. Noah Ngisiang'e - Kenya Marine and Fisheries Research Institute



Figure 22. Participants of the Climate and Physical Processes training track

3.2.3.1 Description of the Study Location

Participants of the Climate and Physical Processes training track undertook field-based learning activities along the Kenyan coast at Pirates Beach (Mombasa) and Gazi Bay (south coast Kenya). These sites were purposefully selected to represent contrasting coastal settings with differing exposure to climate stressors and human influence.

Pirates Beach is an peri-urban area subject to intense anthropogenic pressure, including coastal infrastructure development, shoreline armoring, and high recreational use. In contrast, Gazi Bay is a semi-enclosed coastal embayment dominated by mangrove ecosystems, with relatively lower levels of built infrastructure. The spatial distribution of these sites (Figure 23), enabled participants to compare how physical coastal processes manifest under different environmental and land-use conditions.



Figure 23. A map showing the location of Pirates Beach and Gazi Bay

At Pirates Beach, visible signs of shoreline instability were evident. Figure 24 (a) shows an eroded beach profile characterized by scarping and sediment loss, which provided a clear visual reference for understanding erosion processes driven by wave action, sea-level rise, and human interference. Conversely, Figure 24 (b) presents a mangrove-fringed shoreline in Gazi Bay, where dense vegetation acts as a natural buffer, dissipating wave energy and trapping sediments. This visual contrast was central to discussions on ecosystem-based coastal protection.



Figure 24. Pirates Beach showing erosion features (a), and Gazi Bay (b) with a mangrove-fringed shoreline

At both sites Pyrates Beach and Gazi Bay, participants conducted structured shoreline observations, documented beach profiles, erosion features, and evidence of human modification. Figure 25 captures participants actively engaged in erosion assessment, illustrating practical learning as theoretical concepts related to coastal dynamics applied directly in the field.



Figure 25. Participants conducting shoreline observations along Pirates Beach, Mombasa

At Gazi Bay, participants carried out mangrove species identification exercises to understand the ecological components underpinning physical shoreline stability. Figure 26 shows students identifying dominant mangrove species (*Avicennia marina*, *Rhizophora mucronata*, and *Sonneratia alba*) using morphological features such as root structures and leaf characteristics. This activity reinforced the link between biodiversity, ecosystem function, and coastal resilience.



Figure 26. Mangrove species identification at Gazi Bay, south coast Kenya

To complement field observations, participants analyzed historical sea-level data for Mombasa using datasets from the University of Hawaii Sea Level Center. Data processing and trend analysis were undertaken using Excel and R statistical software.

Both the field and practical sessions enhanced participants understanding of coastal erosion processes, shoreline dynamics, and climate-driven change. Participants gained practical skills in field observation, basic climate data handling, and interpretation of sea-level trends. The resulting sea-level trend (Figure 27), allowed participants to contextualize observed shoreline erosion within broader climate-driven changes. The analysis revealed visible shoreline erosion at both sites and a rising sea-level trend consistent with regional and global climate change patterns. Mangroves and coastal vegetation were shown to play a critical role in reducing erosion, while poorly planned coastal development was observed to increase vulnerability.

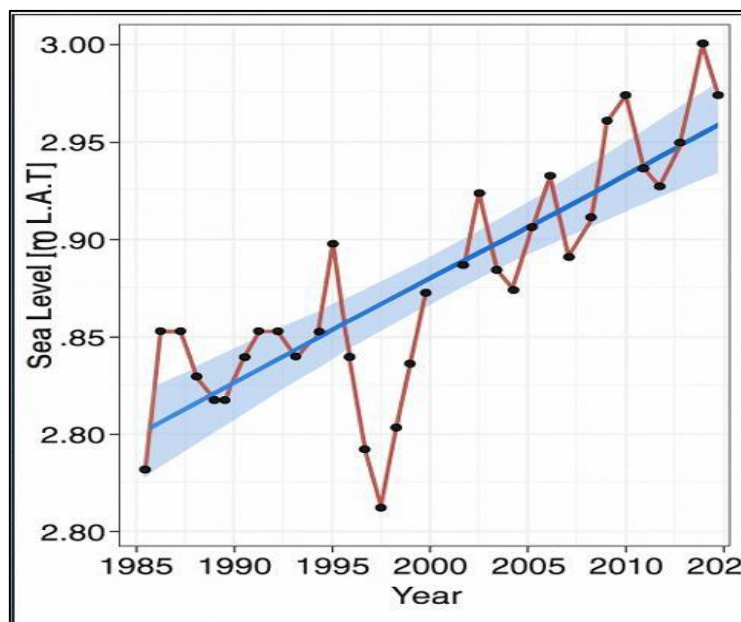


Figure 27. Trend in sea level rise for Mombasa based on historical data

The Climate and Physical Processes group successfully linked theoretical climate concepts with real-world coastal observations and data analysis. The exercise highlighted the importance of ecosystem-based adaptation and informed coastal planning, demonstrating the value of experiential learning for understanding coastal resilience in the WIO region.

3.2.4 Ecological Survey Techniques and Data Management

This training track was facilitated by the following experts:

- Mr. Gabriel Akoko - Early Career Ocean Professionals (ECOP) Program
- Dr. Clara R. Vives - National Aeronautics and Space Administration, USA



Figure 28. Members of the Ecological Survey Technics and Data Analysis group

3.2.4.1 Description of the Study Location

The ecological survey was conducted in Gazi Bay south coast Kenya. This group focused on a mangrove and a seagrass site located approximately 200 m apart. These sites represent a transition from brackish estuarine conditions to more open marine conditions (Figure 29).



Figure 29. A map showing mangrove and seagrass sampling sites, Gazi south coast Kenya

Participants in this training track carried out site characterization using GPS and multi-parameter probes, collected phytoplankton samples, and compiled datasets for analysis. Data cleaning and management were conducted using R statistical software and MS Excel, followed by identification of major phytoplankton groups such as diatoms and dinoflagellates. Exploratory statistical analysis and data visualization were used to assess ecological patterns (Figures 30 and 31).



Figure 30. Sample collection during fieldwork session, Gazi south coast Kenya

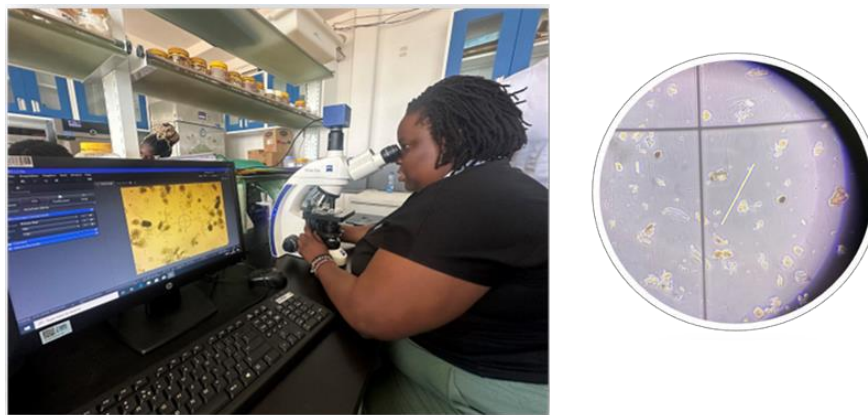


Figure 31. Phytoplankton species identification and characterization at KMFRI Mombasa

Preliminary findings in this group showed higher phytoplankton abundance and diversity at the seagrass site compared to the mangrove site. Diatoms and dinoflagellates were present at both locations, although overall abundance was lower in mangrove areas due to brackish conditions, higher turbidity, and reduced light availability. A potentially toxin-producing dinoflagellate (*Gambierdiscus* sp.) was detected in the mangrove site (Figures 32, 33 and 34).

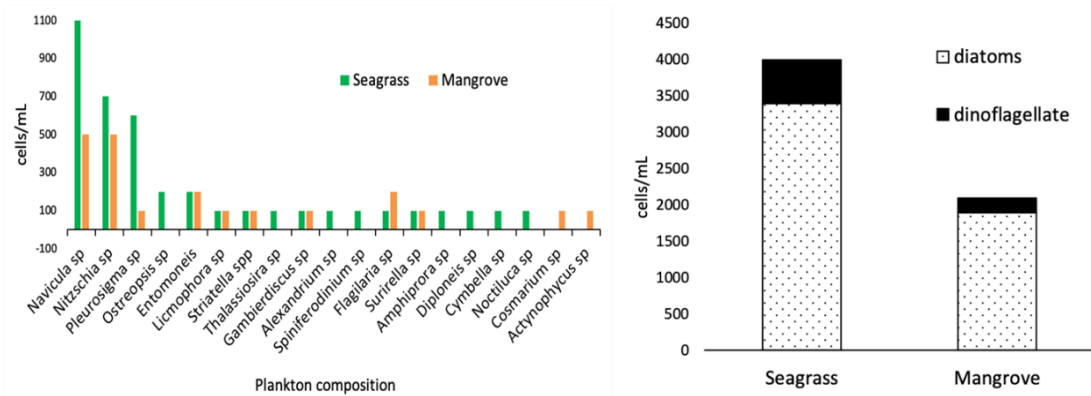


Figure 32. Comparison of phytoplankton diversity and abundance between mangrove and seagrass

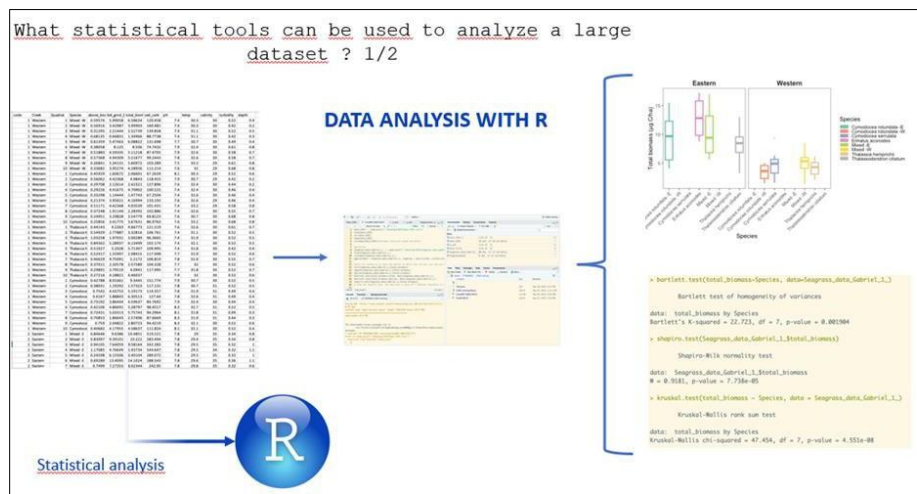


Figure 33. Data analysis using R statistical software

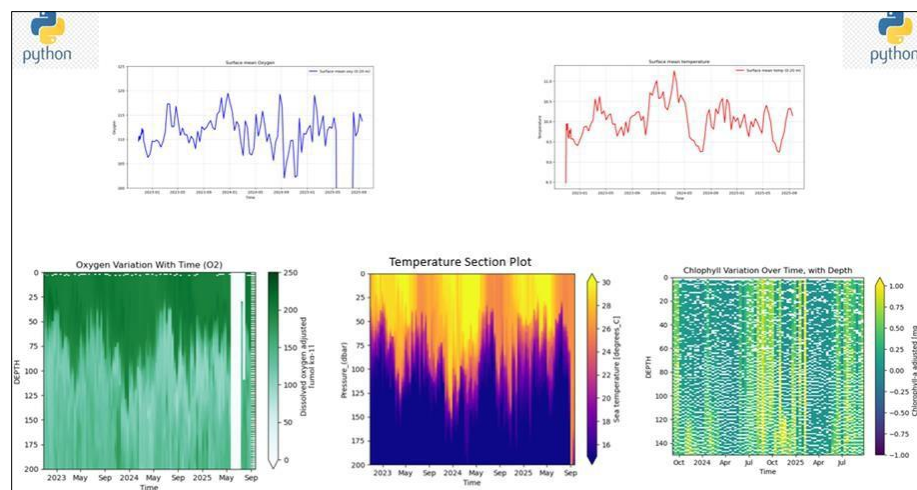


Figure 34. Results of Indian Ocean climatology on Chl-a concentration

The ecological survey demonstrated how phytoplankton communities reflect environmental gradients across coastal habitats. While exploratory in nature, the exercise highlighted the need for replicated sampling and long-term monitoring to strengthen ecosystem assessments.

3.2.5 Marine Pollution and Plastics

This was the training track which attracted the majority of the participants (Figure 35). The following experts facilitated the participants in this training track:

- Dr. Francis Mutuku - Technical University of Mombasa, Kenya
- Prof. Saeed Mwanguni - Technical University of Mombasa, Kenya
- Dr. Janae' Collier - University of Michigan, USA
- Dr. John Mbugani - University of Dodoma, Tanzania
- Dr. John Walter Okwong - University of Lagos, Nigeria
- Dr. Ebenezer Nyadjro - Mississippi State University, USA
- Mr. Richmond Quarcoo - Plastic Punch, Ghana



Figure 35. Participants of the Marine Pollution and Plastics group

3.2.5.1 Description of the Study Location

Fieldwork for this group was conducted at Diani Beach (well managed tourist site) and Shelly Beach (urbanized site) for comparison. These locations were selected to represent contrasting coastal environments influenced by different levels of human activities and waste management practices.

Diani Beach (approximately 4.28° S, 39.59° E) is located south of Mombasa and is known for its sandy shoreline and established tourism infrastructure. Ecologically, it is part of a coastal system that includes coral reefs, seagrass beds, and mangroves, which support

biodiversity and help stabilize the coastline. The area generally benefits from more structured waste management, although tourism-related pressures persist. Shelly Beach (approximately -4.08922, 39.673764 (4° 05' 21" S, 39° 40' 26" E)), located in Mombasa County, represents a more urban-influenced environment with higher population pressure. The site features mixed sandy and rocky shores and is influenced by nearby settlements and drainage systems, leading to increased waste inputs. Coastal processes such as tides and currents contribute to the accumulation of marine debris. Participants also visited black soldier farm where organic matter is recycled to chicken feed and manure (Figures 36 and 37).



Figure 36. Practical field exercises at Shelly and Diani beaches



Figure 37. Organic waste recycling using black soldier fly at HERI-Kenya, south coast

Participants collected beach sediment samples following the COLLECT Protocol and conducted nurdle hunts. Microplastics were extracted using density separation and analysed through electron microscopy at KMFRI laboratories (Figures 38 and 39). Community surveys were conducted to assess awareness of plastic pollution, and a learning visit to a Black Soldier Fly farm explored waste management solutions.

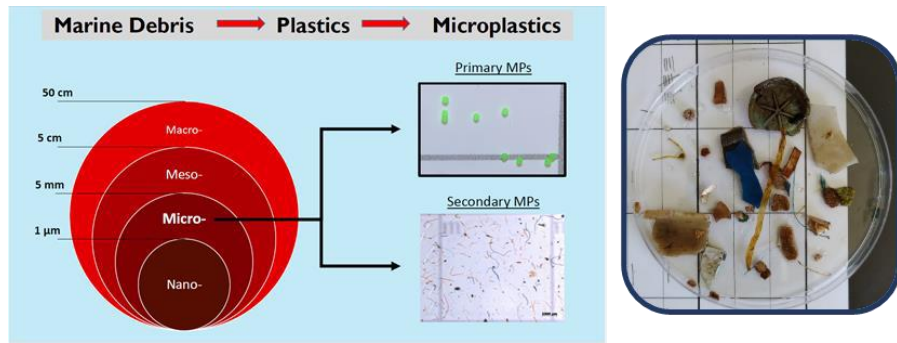


Figure 38. Distribution of microplastic types and colors by site



Figure 39. Analysis of samples at KMFRI labs in Mombasa

Microplastics were detected at both beaches, including visually clean areas. Mwabungu Beach recorded higher microplastic abundance, likely due to greater human activity. Predominant microplastic colours suggested sources such as packaging materials, fishing gear, and consumer products. Community surveys indicated high concern about plastic pollution, particularly among younger respondents. Various findings from this training track are presented in Figures 40, 41 and 42.

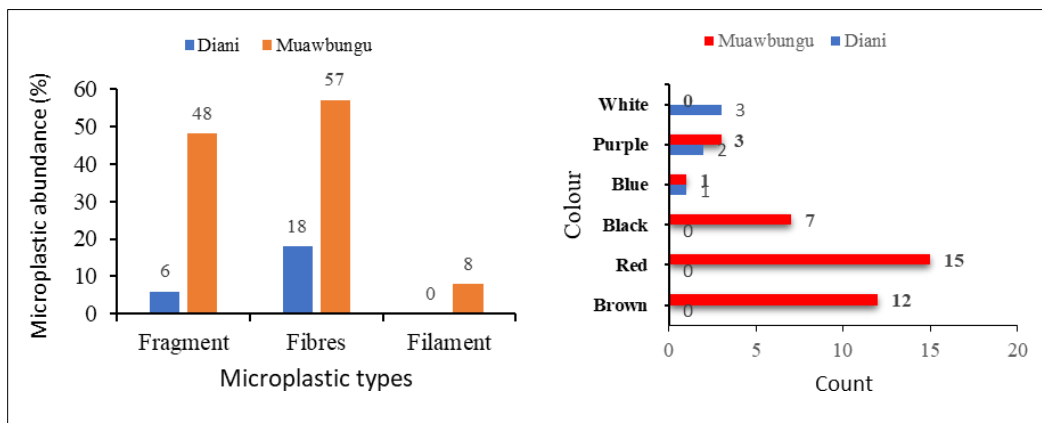


Figure 40. Results of abundance by microplastic types

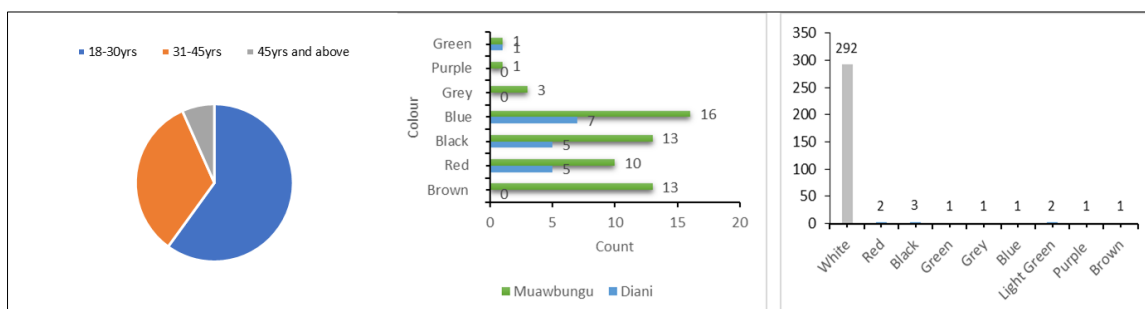


Figure 41. Results of types of microplastics

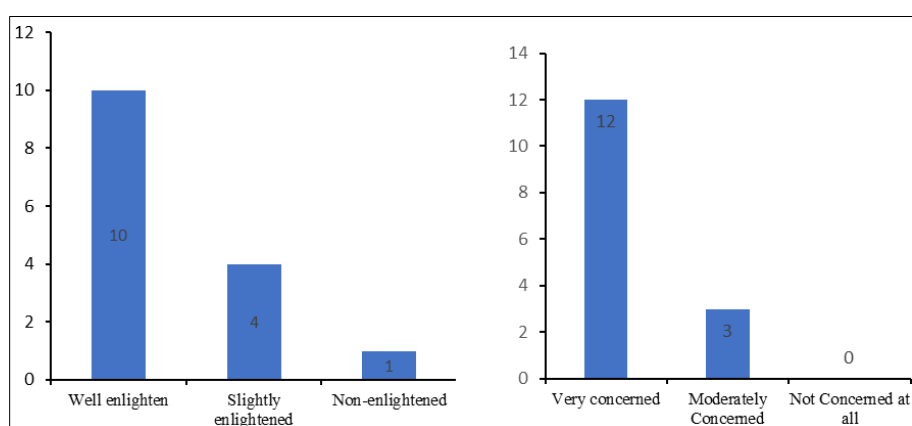


Figure 42. Survey of local community on awareness levels of plastic pollution

The predominance of younger respondents may be due to Diani Beach attracting youth involved in tourism and local businesses, suggesting that awareness campaigns could be more effective if targeted at this group. However, the limited representation of older individuals restricts insights into intergenerational perspectives. While most respondents considered themselves well-informed, indicating prior exposure to information on plastic pollution, some knowledge gaps remain. The high level of concern reflects strong recognition of plastic pollution as a pressing issue, and this awareness could be leveraged to promote community action, though sustained impact that will require behavior change.

The Marine Pollution and Plastics group highlighted the hidden nature of microplastic pollution and demonstrated the value of integrating field sampling, laboratory analysis, and community engagement. The findings reinforce the need for continued monitoring, improved waste management, and preventative measures to protect coastal ecosystems.

3.3 Challenges and Lessons Learned

3.3.1 Logistical and Operational Challenges

1. Large number of applicants

Response of over ten times the targeted number of applications extended the participant selection and verification process beyond the planned timeline, delaying issuance of final admission confirmations.

2. Flight cost fluctuations

Additional travel costs and budget reallocation due to changes in flight ticket costs.

3. Internet bandwidth limitations

During selected training sessions, internet bandwidth at the training venue dropped below functional levels for virtual demonstrations, affecting training activities during the period.

4. Traffic-related movement delays

Peak-hour traffic between TUM and KMFRI caused delays, slightly affecting the start times of scheduled sessions.

3.3.2 Content-Related Challenges

5. Change of training tracks by participants

Some participants requested changes in training tracks, requiring reorganization of practical groups and adjustment of facilitation plans.

6. Trainer unavailability

For the Climate and Physical Processes project track, assigned trainers were in most instances unavailable which resulted in rescheduling and temporary coverage adjustments to ensure continuity of learning outcomes.

3.3.3 Coordination and Administrative Challenge

7. Change of field sites

Change of original field sites due to a travel advisory necessitated identification and confirmation of alternative field sites and revision of logistics and schedules accordingly.

3.4 Feedback from Participants

Feedback was collected from 62 participants through a post-program survey to capture reflections on learning, programme delivery, collaboration, and overall experience. Responses included both structured questions and open-ended comments.

Overall, the feedback showed that the program was positively received, with participants highlighting strong learning value, effective collaboration, and high levels of motivation. Areas for improvement were mainly related to time allocation and program coordination.

3.4.1 Learning Experience and Skills Gained

Participants reported gaining new knowledge and skills, particularly in technical and analytical areas relevant to ocean and coastal science. Hands-on sessions were frequently described as effective in enhancing understanding and linking theory to practice. Many respondents indicated that the program introduced them to tools and concepts they had not previously encountered and strengthened their confidence in engaging with interdisciplinary ocean science topics (Table 1).

Table 1. Summary of learning-related feedback

Aspect	Summary of Responses
Skills gained	Participants reported gaining new technical and analytical skills
Practical learning	Hands-on sessions widely viewed as effective
Relevance	Content considered relevant to academic and professional interests
Application	Improved understanding of real-world ocean science challenges

3.4.2 Program Structure and Time Allocation

A recurring theme across responses was **limited time**, particularly for technical sessions such as programming and data analysis. Participants felt that some sessions were rushed and would benefit from deeper engagement. Scheduling and session flow were also mentioned, with respondents noting that clearer planning and improved time management would enhance learning outcomes (Table 2).

Table 2. Feedback on program structure

Aspect	Summary of Responses
Time allocation	Frequently identified as insufficient
Technical sessions	Participants requested more time and depth
Program flow	Some challenges related to scheduling and pacing
Overall structure	Generally acceptable but could be improved

3.4.3 Group Work and Collaboration

Group-based activities were consistently highlighted as a strong component of the program. Participants valued the opportunity to work in diverse teams, learn from peers, and engage across disciplines and institutions. Collaboration was reported to enhance learning, encourage knowledge exchange, and support networking among participants (Table 3).

Table 3. Feedback on group work and collaboration

Aspect	Summary of Responses
Teamwork	Viewed as effective and enriching
Peer learning	Strongly valued by participants
Diversity	Exposure to different backgrounds and disciplines appreciated
Networking	New professional and academic connections formed

3.4.4 Motivation and Engagement

Beyond technical learning, many participants described the program as motivating and inspiring, particularly for students and early-career professionals. Responses reflected increased interest in ocean-related research and a stronger sense of belonging to the ocean science community.

3.4.5 Logistics and Communication

Logistical arrangements were generally considered adequate, though some respondents noted challenges related to communication, scheduling clarity, and coordination. These issues were described as minor constraints rather than significant barriers to participation.

3.4.6 Overall Assessment

Participants overall feedback indicated that the program was **effective, relevant** and **impactful**, with strong strengths in practical learning, collaboration, and participant engagement. Areas identified for improvement relate mainly to time allocation, scheduling, and communication, which can be addressed to further strengthen future editions.

3.5 Media Coverage

The 1st COES-WIO Training 2025 received broad media coverage across multiple platforms, which enhanced its visibility and outreach at both regional and national levels. Coverage spanned television, print media, radio, digital platforms, documentary, and as an article in WIOMSA Newsletter, reflecting strong public and institutional interest in the programme.

On television, the program was featured on Citizen TV, which provided approximately 10 minutes of live coverage during the official opening ceremony. The closing ceremony also received coverage from Citizen TV as well as Majestic Digital TV, further amplifying the program's reach.

In print media, the event was featured in leading national newspapers, including The People Daily and The Star, highlighting key activities and outcomes of the training.

Radio coverage was provided by Pwani FM, Msenangu FM, and Radio Kaya, extending outreach to local and regional audiences and promoting awareness of ocean science initiatives.

A 20-minute documentary capturing the programme was produced by the Technical University of Mombasa's Corporate Communications Department, documenting key activities, participant experiences, and programme highlights.

In addition, the programme was streamed live across the University's official social media platforms, enabling broader engagement with audiences beyond the physical event and increasing digital visibility.

3.6 Way Forward for COES-WIO

Building on the success of the inaugural COES-WIO training, the program will transition into a rotating regional model, with the next edition planned to be hosted in Tanzania. This approach is intended to strengthen regional ownership, expand geographic reach, and ensure broader participation across the WIO region. In parallel, efforts will be made to formally position COES-WIO as a recognized project under the UN Ocean Decade for Science and Sustainable Development. This will enhance the program's

visibility, align it with global ocean science priorities, and create opportunities for collaboration, resource mobilization, and long-term sustainability.

Future iterations of the program will focus on deepening technical training, particularly in data analysis, modelling, and applied ocean science methods, while maintaining the strong emphasis on field-based and experiential learning.

The program will continue to build and support a regional network of early-career ocean professionals, fostering collaboration, mentorship, and knowledge exchange across countries and disciplines. Strengthening partnerships with academic institutions, research organizations, and development partners will remain critical to scaling the initiative and ensuring its long-term impact.

Lessons learned from the inaugural training will inform improvements in program design, coordination, and logistics. Continued investment in communication and outreach will further enhance visibility and position COES-WIO as a leading regional platform for ocean science capacity development in the WIO region.

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LIST OF APPENDICES

Appendix 1. Newspaper Article on 1st COES-WIO training

14

NEWS FEATURE

PEOPLE DAILY / Friday, 3 October, 2025

Court's move to halt police recruitment a blow to youthful job seekers

by Allen Valentine
@PeopleDailyKe

The country's quest to increase the number of policemen and boost security suffered a debilitating blow after the court halted their recruitment drive, leaving thousands of Kenyan youth in limbo.

In a dramatic blow to job seekers across Kenya, a Nairobi court has issued an injunction suspending the recruitment of 10,000 police constables, derailing a process slated to kick off on October and 2025.

The decision, handed down yesterday by Justice Hellen Wasiha, stems from a legal challenge by former Kilome MP Harun Mwarui against the National Police Service Commission (NPSC) over its handling of the exercise, traditionally managed by the National Police Service (NPS).

The suspension will remain in place until the case is heard and determined, plunging thousands of eager young applicants into uncertainty and frustration.

In court filings, the petitioner contends that NPSC's involvement in police recruitment risked politicisation and inefficiency, potentially compromising the integrity of the Police Service.

Over the past weeks, applicants, primarily aged 18 to 28, had mobilised in droves, submitting online applications and preparing for physical tests at designated county centers starting today.

The genesis of the dispute traces back to the NPSC's bold move to spearhead the recruitment, a departure from longstanding NPS mandate. The commission has also been fighting tooth and nail to grab the Ksh 60 billion annual human resource budgetary allocation from NPS.

Critics have argued that the commission has been overstepping its mandate under the 2010 Constitution, which vests primary responsibility for police operations with the Inspector General (IG).

On Wednesday, the Attorney General (AG) threw weight behind the IG, affirming his role as the custodian of the police payroll and operational framework.

The AG's legal endorsement came as the National Treasury rebuffed the NPSC's request to be given the administration of police payroll.

In a response to the request made by NPSC, Permanent Secretary Dr Chris Kiptoo said splitting the budget would violate public finance principles, as personnel expenses for civil servants are tied to the agencies where they serve.

"This proposed transfer would fragment direct costs across two programs and votes, which contradicts the Public Finance Management Act's emphasis on program-based budgeting," Dr Kiptoo wrote to NPSC.

In Nairobi's Mathare slums, a group of 50 hopefuls who had camped out for updates dispersed in dismay yesterday afternoon, their dreams deferred.



The move by a Nairobi Court to halt police recruitment has plunged job seekers into frustration.

NATIONAL POLICE SERVICE COMMISSION

Kenya's blue economy: Big dreams, stone-age tools

Indian Ocean supports about 65 million people and the youth must seize the opportunities

by Reuben Mwamungu
@PeopleDailyKe

Kenya must adopt the use of modern technologies to monitor its ocean ecosystems and safeguard its resources, according to experts.

According to the Technical University of Mombasa (TUM) Vice Chancellor Professor Laila Abubakar, the technologies currently in use are outdated and slowing efforts to unlock the country's blue economy potential.

She spoke during the Coastal Ocean Environment School-West Indian Ocean Training (COES-WIO Training 2025), which brought together 75 young professionals from 17 countries for a five-day training on oceanography.

"One of the biggest challenges we face is the use of outdated technology. This training will equip experts, especially from Kenya, with the latest tools for blue economy exploration," Prof Abubakar said.

She acknowledged that Kenya's ban on single-use plastics had reduced marine pollution and boosted yields but stressed that much more needs to be done.

"The most important thing for us now is to learn from other countries that are doing better than us and have thrived in the blue economy," she added.

TUM Deputy Vice Chancellor in charge of academic research and extension, Prof Peter Gichangi, said youth must seize the opportunities in the sector, noting that the Indian Ocean supports about 65 million people.

"We are also here to ensure young people interact with others so they can learn more in this field," he said.

Prof Brian Arbie, an oceanographer from the University of Michigan, said the programme links global experts with East African practitioners.

Pwani University Vice Chancellor Prof James Kahindi added that the government is collaborating with TUM, Pwani University and Kenya Marine and Fisheries Research Institute (KMFRRI) to roll out a joint oceanography curriculum, with each institution handling specific units.

The first COES-WIO Training, a summer school initiative by the University of Michigan, has previously been conducted in West Africa, including Ghana and Nigeria. The Kenya edition is co-hosted by TUM, Pwani University and KMFRRI. Planning began in 2023 and, though initially meant to target Kenyan participants only, the initiative grew to attract global partners including the Western Indian Ocean Marine Science Association (WIOMSA) and Early Career Ocean Professionals (ECOP).

Capacity building

The summer school is structured around five training tracks: small-scale fisheries, mariculture and post-harvest management; chemical and physical oceanography; climate change and physical processes; marine pollution and plastics; and data and survey design.

These were selected to build capacity among early-career professionals and students in the region. The modules combine interactive class lectures, group exercises, field work and laboratory practical sessions, with selected sites on Kenya's coast offering practical exposure.

Instructors include Prof Arbie (University of Michigan), Prof Eben Nyadjo (Mississippi University), Prof Richmond Qwaku (University of Ghana), Prof Cosmas Munga (TUM), Prof Francis Mutuku (TUM), Prof Saeed Mwagani (TUM), and Prof Ali Marifa (Pwani University), among others.

Despite covering 70 per cent of the Earth's surface, the ocean remains far less understood than land. Beneath the waters lies an intricate landscape of mountains, valleys, trenches and ridges, most of which remain uncharted.

Global voices have underlined the urgency. Jamie McMichael-Phillips, Director of the Nippon Foundation-GEBCO Seabed 2030 Project, speaking in an earlier forum, noted that less than 26 percent of the ocean floor has been mapped. The rest is still a mystery, with scientists relying on satellite estimates instead of accurate bathymetry—the science of measuring and mapping the underwater depths of oceans, lakes, and rivers.

"We rely on satellite altimeter data for rough estimations, but the actual topography is still largely unknown," he said.

The Seabed 2030 initiative has set a goal of mapping the entire ocean by the end of the decade, but progress is slow. For Kenya, still lagging in technological investment, the gap between rhetoric and reality in the blue economy remains glaring.



Kenya must adopt the use of modern technology to unlock the economic benefits of Blue Economy.

FREEPIK

REGIONAL NEWS

KENYA HOSTS FIRST COASTAL OCEAN ENVIRONMENT SCHOOL

By Cosmas Munga

Opening Ceremony of COES-WIO 2025 in Kenya

22nd September 2023 / Technical University of Mombasa, Kenya

The Coastal Ocean Environment School in the Western Indian Ocean (COES-WIO) is a summer school initiated by the University of Michigan building on their experience of running similar programmes in Ghana and Nigeria. The inaugural COES-WIO 2025 school was co-hosted in Kenya by the Technical University of Mombasa (TUM), Pwani University and the Kenya Marine and Fisheries Research Institute (KMFRRI).

Planning began in 2023, led by Prof. Brian Arbic of the University of Michigan, in collaboration with the Kenyan partners. Originally designed as a national programme for Kenyan participants, the school attracted additional partners including WIOMSA, IOC-UNESCO and early-career professionals and evolved into a regional programme drawing participants and instructors from across the WIO region and beyond. Its objectives are to:

- **enhance regional capacity** in ocean science, especially among early-career professionals;
- **provide hands-on, practical training** in key thematic areas of coastal and marine environmental research;
- **foster Interdisciplinary collaboration and networking** among ocean scientists, researchers and policymakers across Africa;
- **promote the exchange of knowledge** and best practices in sustainable marine resource management;
- **attract professionals from across the globe** to work in partnership with the rich network in the WIO region; and
- **strengthen partnerships** among academic institutions and research organizations within the WIO region.

The first COES-WIO summer school had five hands-on themes: small-scale fisheries, mariculture and post-harvest management; chemical and physical oceanography; climate change and physical processes; marine pollution and plastics; and ecological survey techniques and data management. These themes were selected to build capacity in the WIO region among early-career professionals, post-graduate and undergraduate participants.

The week-long summer school trained a total of **75 participants** drawn from across the WIO region and beyond, with **17 countries** represented.

Participants during a field exercise in Kwale

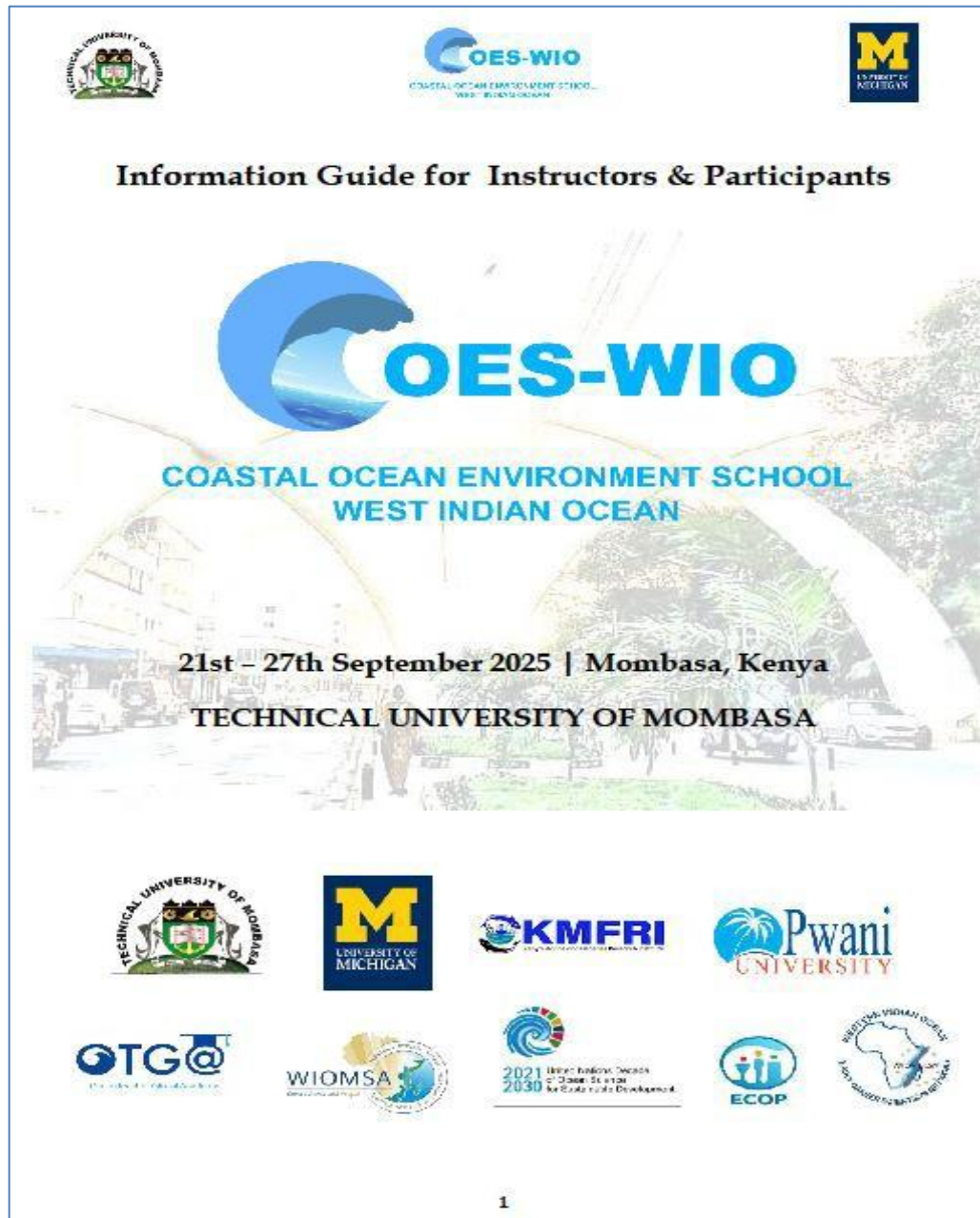
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Appendix 3. Training Program

		
COES-WIO 2025 Coastal Ocean Environment School - Western Indian Ocean 21st - 27th September 2025 Mombasa, Kenya		
Technical University of Mombasa		
Time	Activity	Facilitator/ Moderator
Sunday 21st September 2025		
All Day	Participant arrivals & hotel check-in	All
6:30PM - 9:00PM	Welcome Dinner & Networking	Organizing Committee, TUM
Monday 22nd September 2025		
9:00AM - 11:00AM	Official Opening Ceremony Remarks by: - Prof. Laila Abubakar, VC, TUM - Prof. Joseph Rasowo, DVC, TUM - Prof. Peter Gichangi, DVC (ARE), TUM - Prof. James Kahindi, VC, PU - Dr. Paul Orina, Director General KMFRI - Dr. Janae' Collier, Associate Director, University of Michigan - Dr. Arthur Tuda, Executive Secretary, WIOMSA - Representative, IOC UNESCO - Representative, ECOP	
30 mins	Entertainment	Cultural performance
11:00AM - 11:30AM	Tea/ Coffee Break	All
11:30AM - 11:45AM	History of the School	Prof. Brian Arbic - University of Michigan
11:45AM - 12:30PM	Meet the Facilitators & Track Introductions	Track Leads
12:30PM - 2:00PM	Lunch	All
2:00PM - 5:00PM	Classroom Session 1	Track Facilitators
5:00PM - 5:30PM	Tea/ Coffee Break	All
Tuesday 23rd September 2025		
9:00AM - 12:30PM	Field Excursion 1 - Site visits, data collection, community interviews	Track Facilitators
12:30PM - 2:00PM	Lunch	All
2:00PM - 5:00PM	Field Excursion 1 (continued)	Track Facilitators
5:00PM - 5:30PM	Tea/ Coffee Break	All
Wednesday 24th September 2025		
8:00AM - 2:00PM	Field Excursion 1 (continued)	Track Facilitators
4:00PM - 6:00PM	Group Reflection & Debrief	Facilitators
Evening	Dinner at accommodation	All
Thursday 25th September 2025		
8:00AM - 2:00PM	Lab work & Data Analysis	Track Facilitators
4:00PM - 6:00PM	Classroom Session 4 - Data Analysis & Reporting	All
Evening	Dinner at accommodation	All

		
COES-WIO 2025 Coastal Ocean Environment School - Western Indian Ocean 21st - 27th September 2025 Mombasa, Kenya		
Time	Activity	Facilitator/ Moderator
Friday 26th September 2025		
8:00AM - 10:00AM	Student Presentations	Students
10:00AM - 10:30AM	Tea/ Coffee Break	All
10:30AM - 12:30PM	Closing Ceremony & Feedback Certificate Awards	Organizing Committee
12:30PM - 2:00PM	Lunch	All
2:00PM - 5:00PM	Wrap-up & Graduate School Session	Facilitators
7:00PM - 12:00AM	Farewell Dinner Entertainment - Cultural dance	TUM & Partners
Saturday 27th September 2025		
All Day	Participant check-out & departure	All

Appendix 4. 1st COES-WIO Information guide



Appendix 5. List of Instructors

S/No	First Name	Last Name	Email	Professional Affiliation	Country of Passport
1.	Brian	Arbic	arbic@umich.edu	University of Michigan	USA
2.	Tracey	Schafer	tschafer25@gmail.com	University of Florida	USA
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5.	Ebenezer	Nyadjro	esn31@msstate.edu	Mississippi State University	USA
6.	Clara	R Vives	clararvives@outlook.com	Japan Agency for Marine-Earth Sciences & Technology (JAMSTEC)	Spain
7.	John	Mbugani	johnmbugani@gmail.com; john.mbugani@udom.ac.tz	The University of Dodoma (UDOM)	Tanzania
8.	Frank	Mirobo	frankmirobo.fm@gmail.com	The University of Dodoma (UDOM) & WIO Early Career Scientists Network	Tanzania
10.	Janae'	Colliers	drjanae@umich.edu	University of Michigan	USA
11.	Joseph	Tunje	g.tunje@pu.ac.ke	Pwani University	Kenya
12.	Chengzhu	Xu	chengzhu.xu@oregonstate.edu	Oregon State University	Canada
13.	Luis	Pareja Roman	pareja@marine.rutgers.edu	Rutgers University	Colombia
14.	Stanton	Belford	sbelfor2@utsouthern.edu	University of Tennessee Southern	USA
15.	Audrey	Delpech	audrey.delpech@cnrs.fr	CNRS (Centre National de la Recherche Scientifique)	France
17.	Okwong	Walter	okwong56@gmail.com	Waste Management Association of Nigeria	Nigeria
18.	Richmond	Quarcoo	plasticpunchngo@gmail.com	Plastic Punch	Ghana
19.	Gabriel	Juma		ECOP	Germany
20.	Mariam	Swaleh	mariamsaria@gmail.com	TUM	Kenya
21.	Saeed	Mwaguni	smwaguni@tum.ac.ke	TUM	Kenya
22.	Francis	Mutuku	fmutuku73@gmail.com	TUM	Kenya
23.	Ali	Mwakumanya	m.mwakumanya@pu.ac.ke	Pwani University	Kenya
24.	Amon	Kimeli	akimeli@kmfri.co.ke	KMFRI	Kenya
25.	Noah	Ngisiang'e	ngisiange@gmail.com	KMFRI	Kenya
26.	Peter	Odote		KMFRI	Kenya
27.	Cosmas	Munga	Ckamunga2014@tum.ac.ke	TUM	Kenya

Appendix 6. List of participants

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Physical and Chemical Oceanography				
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Ecological survey Techniques and Data Analysis				
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